

Defect-Tolerant Design of Lead-Free RbGeI₃ Solar Cells: From SCAPS-1D Optimization to Manufacturing Tolerances, Benchmarked Against CsGeI₃

Md. Abdul Malek Fahim^{a,1}, Hussain Touhid Siddiquee^{a,2,*}

^a*Department of Electrical and Electronic Engineering, Leading University, Sylhet,
Bangladesh*

Abstract

The rapid rise of lead-halide perovskite solar cells has been tempered by lead toxicity, motivating a search for lead-free absorbers with comparable optoelectronics. Rubidium germanium iodide (RbGeI₃) offers a direct, tunable bandgap, strong absorption, and improved environmental compatibility. Here we report a SCAPS-1D (v3.3.10) optimization of an RbGeI₃ cell. After screening eight ETL–HTL combinations sharing the same absorber, the FTO/TiO₂/RbGeI₃/CuI/Au architecture was carried forward on stability and cost grounds. Eleven parameters (absorber thickness, defect density, dielectric constant, electron affinity, bandgap, effective densities of states, transport-layer thicknesses, both interface defect densities) were varied sequentially under AM 1.5G at 300 K. The optimized device reaches 26.81% PCE, with $V_{OC} = 1.13$ V, $J_{SC} = 30.32$ mA/cm², FF = 78.26%: a 700 nm absorber ($E_g = 1.4$ eV, $\chi = 3.9$ eV, $\epsilon_r = 15$, $N_C = N_V = 1 \times 10^{17}$ cm⁻³), 10 nm TiO₂, 100 nm CuI, and 1×10^{14} cm⁻² interface defects. The band diagram confirms favorable alignment: a 1.8 eV offset at CuI/RbGeI₃ blocking electron back-transfer and a 0.10 eV step at RbGeI₃/TiO₂ aiding extraction. The 26.81% is an ideal-coupling upper bound; optical and defect realism bound the plausible window to ≈ 20.7 –23.5%, and joint optimization places the mathematical maximum at 31.78%.

*Corresponding author.

¹E-mail: mdabdulmalekfahim@gmail.com; fahim.eee@lus.ac.bd

²E-mail: touhidsiddiqueeraj@gmail.com

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I. Introduction

I.A. Background and motivation

Global electricity demand continues to climb as populations grow, economies industrialize, and urbanization accelerates. Fossil fuels still supply the bulk of this demand [1, 2], but their depletion and the environmental cost of carbon emissions have made renewable energy a strategic priority rather than an alternative. Among renewable sources, solar energy stands out for its abundance and for the steady decline in the levelized cost of photovoltaic (PV) electricity, which has made solar the fastest-expanding renewable technology worldwide [3]. The global installed PV capacity surpassed 2.2 TW in 2024, and continued growth relies on identifying photovoltaic materials that combine high efficiency with low environmental impact and low fabrication cost. Continued progress in PV technology matters for meeting climate-mitigation targets and for expanding energy access in regions without reliable grid infrastructure [4]. Photovoltaic devices are commonly classified into three generations: crystalline silicon (high efficiency, energy-intensive), thin films (less material, modest efficiency), and emerging devices promising high efficiency at low processing cost.

I.B. Perovskite solar cells and the lead problem

Halide perovskites adopt an ABX₃ crystal structure in which the A-site hosts a monovalent cation, the B-site a divalent metal, and the X-site a halogen. Their attractiveness for photovoltaics stems from strong light absorption, continuously tunable bandgaps, ambipolar carrier transport with long diffusion lengths, and weak exciton binding [7, 8]. Certified power conversion efficiencies (PCEs) for single-junction lead-based devices now exceed 26% [3], and perovskite-on-silicon tandems have crossed 33% [3]. The best absorbers, however, are methylammonium lead iodide (MAPbI₃) and formamidinium lead iodide (FAPbI₃), both of

which contain lead. Lead contamination concerns across the device life cycle drive the search for Sn, Ge, Bi, and Sb substitutes [9, 10, 11, 12]. Tin-based perovskites initially attracted the most attention as lead substitutes, because Sn^{2+} has a similar lone-pair electronic configuration to Pb^{2+} [11]. Unfortunately, Sn^{2+} readily oxidizes to Sn^{4+} in air, generating high densities of deep defects and accelerating device degradation [11, 12]. Germanium-based perovskites have emerged as a complementary alternative: germanium offers the same lone-pair electronic structure as lead, [13]. Among Ge-based absorbers, rubidium germanium iodide (RbGeI_3) stands out for its suitable bandgap (≈ 1.3 – 1.4 eV depending on the structural phase and computational method), strong optical absorption coefficient, and good thermal stability [14, 15, 16]. DFT confirms a direct gap with MAPbI_3 -comparable absorption [13, 17, 18].

I.C. Role of numerical simulation and research gap

Because experimental optimization of a multilayer photovoltaic device is both time-consuming and expensive, numerical simulation has become the standard tool for screening materials and parameters before fabrication. The Solar Cell Capacitance Simulator in one dimension (SCAPS-1D), developed at the Department of Electronics and Information Systems (ELIS) of Ghent University, Belgium, has been widely adopted for this purpose [19, 20, 21]. SCAPS-1D solves Poisson, continuity, and drift-diffusion equations self-consistently [19], disentangling individual parameter contributions (Figure 1).

Although several SCAPS-1D studies of RbGeI_3 devices have been reported, most have optimized only a limited subset of device parameters or investigated a single device configuration. The simulation baseline for RbGeI_3 devices is still low: Pindolia et al. reported a PCE of 9.77% for the $\text{FTO}/\text{TiO}_2/\text{RbGeI}_3/\text{NiO}/\text{Ag}$ architecture using SCAPS-1D simulation [14]. That study swept HTLs, ETLs, thicknesses, doping, defects, contacts, and temperature broadly [14]. Simulation-based studies have since raised the predicted PCE of RbGeI_3 devices to 27.21% (2024 [22]) and 26.47% (2026 [23]), although each of these works optimized only a limited subset of device parameters. The ETL–HTL

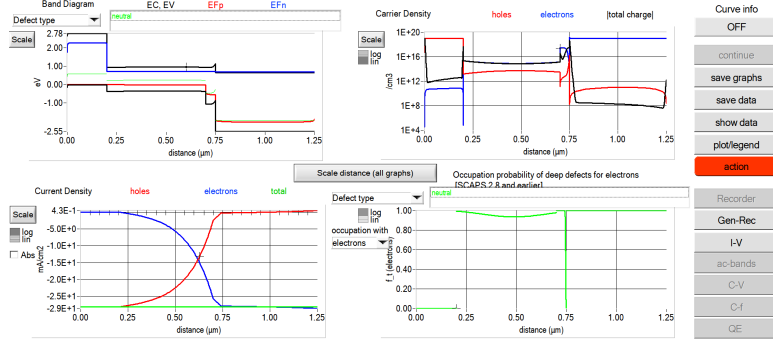


Figure 1: Graphical user interface of SCAPS-1D (version 3.3.10) used for device simulation in this study.

combinations explored for RbGeI_3 have also been narrow: most studies used TiO_2 as the ETL and either Spiro-OMeTAD, NiO , or CBTS as the HTL, while the TiO_2/CuI combination used here has received comparatively little attention despite the well-documented advantages of CuI as an HTL.

I.D. Contributions and paper organization

We close this gap by running a SCAPS-1D optimization of an FTO/ TiO_2 / RbGeI_3 / CuI /Au perovskite solar cell. We first screen eight ETL-HTL combinations, then run an eleven-step sequential optimization of absorber, interface, and transport-layer parameters. We also analyze spectra, band diagram, and primary-source-verified literature benchmarks.

II. Literature Review and Research Gap

Over the past five years, SCAPS-1D, often combined with first-principles DFT calculations, has become the workhorse numerical tool for designing and optimizing lead-free perovskite solar cells before experimental fabrication. This section surveys the most relevant recent SCAPS-1D studies on Ge-based and Sn-based absorbers, highlighting structure, performance, and limitations. Supplementary Table S1 summarizes these studies. All numerical values listed in

Supplementary Table S1 have been cross-checked against the primary source literature; where a reported efficiency was inconsistent with the device’s published J–V metrics, the value implied by those metrics was adopted.

Copper(I) iodide is a wide-gap ($E_g \approx 3.1$ eV), p-type transparent conductor whose valence-band maximum aligns closely with that of RbGeI₃, supporting efficient hole extraction [25, 26]. Reported single-crystal hole mobilities of CuI exceed $100 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, and the material is solution-processable at low temperatures, which suits large-area fabrication [27]. Despite these advantages, only a few SCAPS-1D studies have paired CuI with RbGeI₃, and none has run an eleven-parameter optimization on this combination. Here we vary all eleven device-level parameters on the FTO/TiO₂/RbGeI₃/CuI/Au architecture and report the resulting photovoltaic performance. A further observation from Supplementary Table S1 is that the comparator device structures span a range of ETL and HTL materials (TiO₂, PCBM, C₆₀, NiO, CuI, CBTS, Spiro-OMeTAD, PEDOT:PSS) and back-contact metals (Ag, Au). Two details prevent citation-propagation errors: the Loumachi ETL is C₆₀ (not “C6”) and its back contact is Ag, not Au [22].

III. Device Structure and Material Selection

III.A. Proposed device architecture

The lead-free perovskite solar cell we propose uses a conventional n-i-p planar heterojunction architecture with the configuration FTO/TiO₂/RbGeI₃/CuI/Au. The device comprises five functional layers: a fluorine-doped tin oxide (FTO) transparent conducting oxide front electrode, a titanium dioxide (TiO₂) electron transport layer (ETL), a rubidium germanium iodide (RbGeI₃) absorber layer, a copper iodide (CuI) hole transport layer (HTL), and a gold (Au) back contact. This stack combines efficient charge separation, rapid carrier transport, and suitable energy-level alignment while eliminating the environmental and health concerns associated with lead-based perovskite absorbers. The schematic diagram of the proposed device structure is shown in Figure

2. The choice of constituent materials is justified in the remainder of this section, and the initial material parameters used in the SCAPS-1D simulation are summarized in Supplementary Table S2.

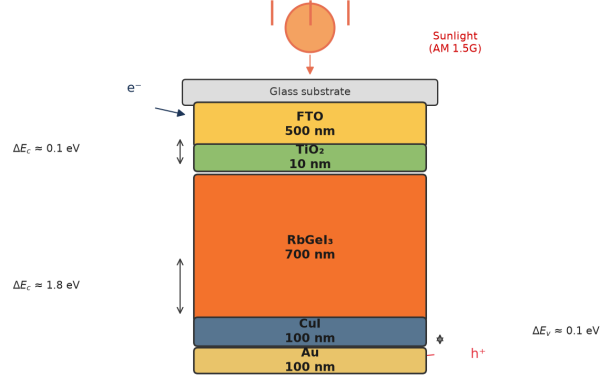


Figure 2: Schematic diagram of the proposed FTO/TiO₂/RbGeI₃/CuI/Au perovskite solar cell structure, showing the five functional layers and the direction of photogenerated carrier transport under AM 1.5G illumination.

III.B. Material selection rationale

Fluorine-doped tin oxide (FTO, 500 nm) is the front electrode for its transparency, conductivity, and thermal/chemical stability [28]. It was preferred over ITO for stability, process compatibility, and cost [29]. Titanium dioxide (TiO₂) is the electron transport layer: its 3.2 eV gap passes visible light, its conduction-band minimum lies slightly below typical iodide-perovskite levels for downhill electron transfer, and its deep valence band blocks holes [30, 31]. Despite lower mobility than ZnO/SnO₂, it transports adequately and forms stable perovskite interfaces [32]. Rubidium germanium iodide (RbGeI₃, ABX₃) is the absorber: replacing lead with germanium keeps the needed optoelectronics, with a direct 1.3–1.4 eV gap and strong absorption (comparable to MAPbI₃ by first-principles calculations [17, 18]), and good thermal stability make it well-suited for single-junction photovoltaic applications. Screening used 400 nm and

1.4 eV from experimental literature [14]. Both parameters were subsequently optimized (Section V). Copper iodide (CuI) is the hole transport layer for its conductivity, transparency, close valence alignment with RbGeI₃, and low cost [25, 26, 33]. CuI is a wide-gap ($E_g \approx 3.1$ eV) p-type conductor whose valence maximum aligns with RbGeI₃ (0.10 eV offset, Supplementary Section S1.3); single-crystal hole mobilities exceed $100 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and it processes from solution at low temperature [27]. Screening used 100 nm CuI. Gold (Au) is the back contact: its high work function gives an ohmic contact to CuI without a parasitic Schottky barrier [34].

III.C. Initial material parameters

The initial electrical and optical parameters used in the SCAPS-1D simulation of the proposed FTO/TiO₂/RbGeI₃/CuI/Au solar cell are summarized in Supplementary Table S2. These values were collected from previously published experimental and numerical studies on lead-free perovskite solar cells [14, 15, 16, 19, 30] and were carefully incorporated into the SCAPS-1D simulator to establish a realistic baseline device before performing further optimization. The interface defect densities applied at the two heterojunctions are listed in Supplementary Table S3. The initial absorber thickness of 400 nm listed in Supplementary Table S2 is the value at which all eight device configurations were screened (Section V.A, Table I). This thickness was subsequently optimized to 700 nm in the consolidated final device (Section V.G).

IV. Methodology

IV.A. SCAPS-1D simulation framework

The device stack was simulated with SCAPS-1D version 3.3.10 [19, 20, 21], which solves the Poisson equation together with the electron and hole continuity equations self-consistently across the device and is widely used for thin-film and perovskite device studies. The drift-diffusion transport model underlying

these equations is summarized below; further details are available in the original references [19, 20, 21].

$$\frac{d}{dx} \left(\epsilon \frac{d\psi}{dx} \right) = -q (p - n + N_D - N_A + \rho_t) \quad (1)$$

$$\frac{dJ_n}{dx} = q (R - G) \quad (2)$$

$$\frac{dJ_p}{dx} = q (G - R) \quad (3)$$

where ϵ is the dielectric permittivity, ψ the electrostatic potential, q the elementary charge, n and p the electron and hole concentrations, N_D and N_A the donor and acceptor concentrations, and ρ_t the trapped charge density. Electron and hole transport are governed by the continuity equations, where J_n is the electron current density, G the carrier generation rate, and R the carrier recombination rate; the hole continuity equation describes hole transport analogously, where J_p is the hole current density. Together, Eqs. (1)–(3) describe generation, transport, and recombination under illumination; SCAPS-1D computes the resulting J–V characteristics from the layer properties defined below.

IV.B. Simulation conditions

All simulations were performed under the AM 1.5G spectrum at an incident power density of 100 mW/cm² (one sun, STC) with the operating temperature fixed at 300 K, using the initial material parameters of Supplementary Table S2 and the interface defect densities of Supplementary Table S3 unless otherwise stated. During each optimization step, only one parameter was varied while all remaining parameters were held constant; the general simulation conditions are summarized in Supplementary Table S4. Absorption uses the SCAPS square-root law $\alpha = A\sqrt{h\nu - E_g}$ with $A = 10^5 \text{ cm}^{-1}$ (uniform at the electrical bandgap). Contacts are ideal ($R_s = 0$, $R_{sh} = \infty$); no series/shunt resistance is included unless stated, so fill factors are upper bounds. Figures 3–13 record each sweep as-run with its contemporary deck version; absolute offsets between sweep generations (e.g. the Section V.B sweep record) are bounded by consolidated-deck re-simulation, and all headline metrics come solely from single consolidated-deck runs.

IV.C. Performance parameters

Performance was evaluated using the four key parameters extracted from the simulated J–V curve: open-circuit voltage (V_{OC}), short-circuit current density (J_{SC}), fill factor (FF), and power conversion efficiency (PCE), computed as $PCE = (V_{OC} \times J_{SC} \times FF)/P_{in}$ with $P_{in} = 100 \text{ mW/cm}^2$ under AM 1.5G.

IV.D. Sequential optimization procedure

Starting from the initial device, one parameter at a time (OPAT) was varied while all others were held constant, as detailed in Section V. The procedure comprises eleven optimization steps: (1) absorber thickness ($0.3\text{--}1.0 \mu\text{m}$); (2) absorber bulk defect density ($10^{12}\text{--}10^{18} \text{ cm}^{-3}$); (3) absorber dielectric constant ($15\text{--}23.1$); (4) absorber electron affinity ($3.9\text{--}4.2 \text{ eV}$); (5) $\text{TiO}_2/\text{RbGeI}_3$ interfacial defect density ($10^{12}\text{--}10^{18} \text{ cm}^{-2}$); (6) $\text{RbGeI}_3/\text{CuI}$ interfacial defect density ($10^{12}\text{--}10^{18} \text{ cm}^{-2}$); (7) absorber bandgap ($1.3\text{--}1.7 \text{ eV}$); (8) TiO_2 ETL thickness ($0.01\text{--}0.09 \mu\text{m}$); (9) CuI HTL thickness ($0.1\text{--}1.9 \mu\text{m}$); (10) conduction-band effective density of states N_C ($1 \times 10^{17}\text{--}1 \times 10^{20} \text{ cm}^{-3}$); (11) valence-band effective density of states N_V ($1 \times 10^{17}\text{--}1 \times 10^{20} \text{ cm}^{-3}$). Each step’s locally optimal parameter value was carried forward into all subsequent steps, with the consolidated final parameter set summarized in Table III. Because each per-step optimum is identified with all other parameters fixed at prior-step values, the consolidated parameter set may not be a global optimum; the OPAT approach nevertheless identifies the dominant parameter effects and establishes a clear baseline. To test for missed cross-interactions, Section V.G reports a genetic-algorithm (GA) joint optimization over the six most sensitive parameters (bandgap, bulk and interface defect densities, N_C , N_V , absorber thickness), which finds a higher mathematical optimum at a lower bandgap and lower defect density than the OPAT point.

IV.E. Device configurations screened

Eight planar configurations combining two ETLs (TiO_2 , PCBM) with four HTLs (NiO , CuI , CBTS, Spiro-OMeTAD) around the RbGeI_3 absorber (D1–D8,

Table I) were simulated at the initial absorber thickness of 400 nm (Supplementary Table S2) to permit a fair architecture comparison before the thickness optimization of Section V.B.

Device	Architecture
D1	FTO / PCBM / RbGeI ₃ / NiO / Au
D2	FTO / TiO ₂ / RbGeI ₃ / NiO / Au
D3	FTO / PCBM / RbGeI ₃ / CuI / Au
D4	FTO / TiO ₂ / RbGeI ₃ / CuI / Au
D5	FTO / PCBM / RbGeI ₃ / CBTS / Au
D6	FTO / TiO ₂ / RbGeI ₃ / CBTS / Au
D7	FTO / PCBM / RbGeI ₃ / Spiro-OMeTAD / Au
D8	FTO / TiO ₂ / RbGeI ₃ / Spiro-OMeTAD / Au

Table I: Eight device configurations screened in this study. All configurations were simulated at an absorber thickness of 400 nm.

V. Results and Discussion

V.A. Screening of device configurations

Table II compiles the photovoltaics of all eight configurations: two ETLs (TiO₂, PCBM) crossed with four HTLs (NiO, CuI, CBTS, Spiro-OMeTAD) on the common RbGeI₃ absorber. The D4 architecture (TiO₂/CuI) combines well-matched band alignment with an all-inorganic transport stack, avoiding the stability limitations of organic HTLs.

The best FF belongs to D7 (Spiro-OMeTAD, 85.01%) and the best PCE to D2 (21.10%), not to D4. D4 still won our attention: CuI beats Spiro-OMeTAD on stability and cost and NiO on moisture and oxygen sensitivity while offering higher hole mobility. Its PCE deficit of about 1.3 percentage points to the top entries is the kind of gap the parameter sweep can close; the stability and cost that CuI carries are not something parameter tuning can buy back.

Dev.	Architecture	V_{OC} (V)	J_{SC} (mA/cm ²)	FF (%)	PCE (%)
D1	FTO/PCBM/ RbGeI ₃ /NiO/Au	0.81	30.10	83.68	20.40
D2	FTO/ TiO ₂ /RbGeI ₃ /NiO/ Au	0.82	30.46	84.48	21.10
D3	FTO/PCBM/ RbGeI ₃ /CuI/Au	0.81	30.10	83.96	20.47
D4	FTO/ TiO ₂ /RbGeI ₃ /CuI/ Au	0.812	30.46	79.98	19.78
D5	FTO/PCBM/ RbGeI ₃ /CBTS/ Au	0.82	30.20	82.36	20.40
D6	FTO/ TiO ₂ /RbGeI ₃ /CBTS/ Au	0.82	30.55	82.13	20.57
D7	FTO/PCBM/ RbGeI ₃ /Spiro/Au	0.81	30.07	85.01	20.71
D8	FTO/ TiO ₂ /RbGeI ₃ /Spiro/Au	0.82	30.46	83.95	20.97

Table II: Initial photovoltaic performance of the eight screened device configurations, simulated at an absorber thickness of 400 nm. Selected as the baseline for full parametric optimization.

V.B. Optimization summary

Eleven parameters were optimized one at a time on the screened FTO/TiO₂/RbGeI₃/CuI/Au stack (absorber thickness, bulk and interface defect densities, dielectric constant, electron affinity, bandgap, transport-layer thicknesses, conduction- and valence-band densities of states); each optimum was carried forward, giving the consolidated set in Table III. Full per-parameter traces (Figs. S1–S10) are given in Supplementary Section S1, and the illumination and dark characteristics (Figs. S11–S12) in Section S2. The defect-density step itself is retained below (Section V.C) together with its energy/asymmetry extension, since Ge-related degradation is central to this work.

V.C. Effect of absorber defect density

Bulk defects act as SRH recombination centers: swept from 10^{12} to 10^{18} cm⁻³, the device holds to 10^{14} cm⁻³ without loss, then PCE collapses from 20.40% to 7.58% at 10^{18} cm⁻³ through falling V_{OC} (0.848 to 0.550 V). The sweep settles at 10^{14} cm⁻³, the standard lead-free assumption. Beyond the total trap density, the energetic position E_t of the dominant defect and the electron/hole capture asymmetry (σ_n , σ_p) were swept to model Ge-related degradation explicitly. At $N_t = 10^{15}$ cm⁻³, moving E_t from 0.2 eV to 1.2 eV above E_v traces a shallow symmetric minimum centred at mid-gap: PCE falls from 25.01% at the band edges to 24.84% for $E_t = 0.4$ -1.0 eV (Fig. 12), the textbook SRH fingerprint that mid-gap states are the most lethal recombination centres. At the baseline $N_t = 10^{14}$ cm⁻³ the level position is immaterial (26.81-26.85%), confirming the consolidated defect budget sits deep inside the tolerant regime. Capture asymmetry at mid-gap (Fig. 13) is significant even at a fixed $\sigma_n \times \sigma_p$ product: the electron-capture-dominated trap ($\sigma_n = 10^{-17}$, $\sigma_p = 10^{-19}$ m²) limits PCE to 23.97%, while the mirrored hole-dominated trap holds 26.71%, and the symmetric $10^{-17}/10^{-17}$ case collapses to 21.45%. The oxidation threat ($\text{Ge}^{2+} \rightarrow \text{Ge}^{4+}$, Ge vacancies) therefore cannot be reduced to a single N_t number: its energy and capture character jointly set the damage, and dense symmetric traps are the worst case.

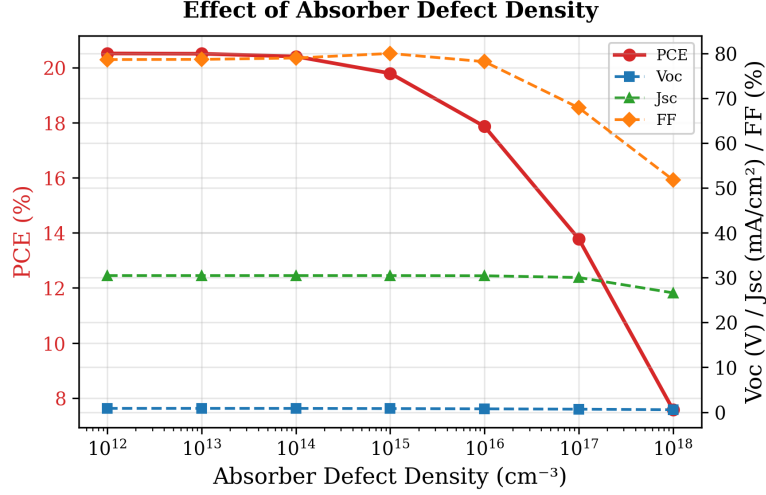


Figure 3: Variation of PCE, V_{OC} , J_{SC} , and FF with $RbGeI_3$ absorber defect density. Performance degrades sharply above 10^{14} cm^{-3} .

V.D. Quantum efficiency analysis

Figure 4 tracks the quantum efficiency: QE climbs from 34.52% at 300 nm to near 100% at 400 nm, the suppressed ultraviolet response being surface recombination of near-front absorption [21, 38]. Absorption in the visible is excellent: QE holds near 100% from 400 to 650 nm, then eases to 76% at 850 nm and cuts off near 900 nm, matching the 1.4 eV gap ($\lambda_c \approx 886 \text{ nm}$). Integrating the spectrum gives $\approx 23.3 \text{ mA/cm}^2$, short of the 30.32 mA/cm^2 read from the J–V curve. The shortfall is a deck-version difference, not physics: the QE spectrum comes from the earlier sweep deck while J–V uses the consolidated deck (flat/graded endpoints give identical J–V, $\Delta J_{SC} = 0.00$). The QE curve is therefore retained only for its edge shape ($\lambda_c \approx 886 \text{ nm}$, consistent with 1.4 eV); the J–V-derived $J_{SC} = 30.32 \text{ mA/cm}^2$ is the reported metric.

V.E. Energy band diagram analysis

The energy band diagram of the optimized device under illumination (Figure 5) shows clean extraction paths at both heterojunctions; both interfaces are thus electronically well passivated. From affinities ($\chi = 2.1$ vs 3.9 eV) the conduction

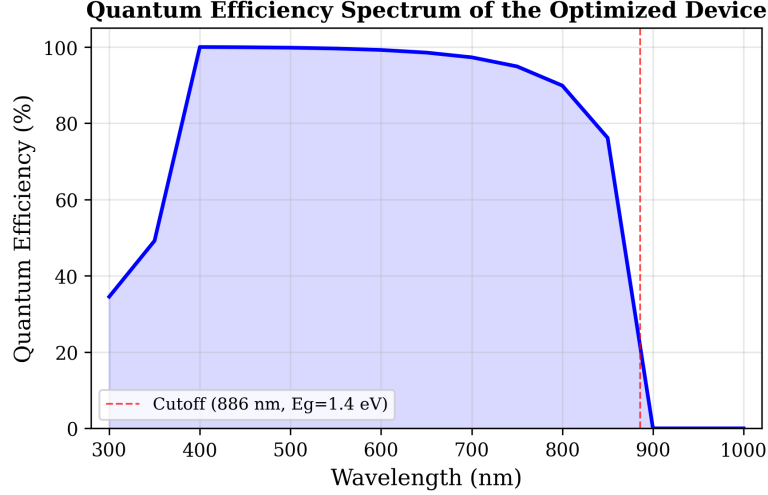


Figure 4: Variation of quantum efficiency (QE) with incident wavelength of the optimized solar cell. The cutoff at ≈ 886 nm corresponds to the absorber bandgap of 1.4 eV.

offset is $\Delta E_c \approx 1.8$ eV (electron-blocking), while $E_v = -(E_g + \chi)$ gives $\Delta E_v \approx 0.10$ eV, so holes cross freely; both band edges stay flat across the 700 nm absorber, consistent with the 1.4 eV gap near the Shockley–Queisser optimum. At $\text{RbGeI}_3/\text{TiO}_2$ ($\chi = 3.9$ vs 4.0 eV, ≈ 800 nm from the back contact) a 0.10 eV downward conduction step assists extraction while a 1.90 eV valence offset ($E_v = -5.30$ vs -7.2 eV) blocks holes. Under illumination the quasi-Fermi levels split, F_n tracking the conduction band and F_p the valence band; the splitting sets $V_{OC} \approx 1.13$ V, with slight bending at the heterojunctions from the built-in field.

V.F. Current density–voltage characteristics

The current density–voltage (J–V) characteristic of the consolidated optimized device under AM 1.5G illumination is shown in Figure 6. The curve is sharply rectangular, with an open-circuit voltage of $V_{OC} = 1.13$ V, a short-circuit current density of $J_{SC} = 30.32$ mA/cm², and a fill factor of 78.26%, yielding a PCE of 26.81%. The maximum-power point occurs near $V = 0.96$ V with $J \approx 27.9$ mA/cm², where the instantaneous power density reaches 26.8

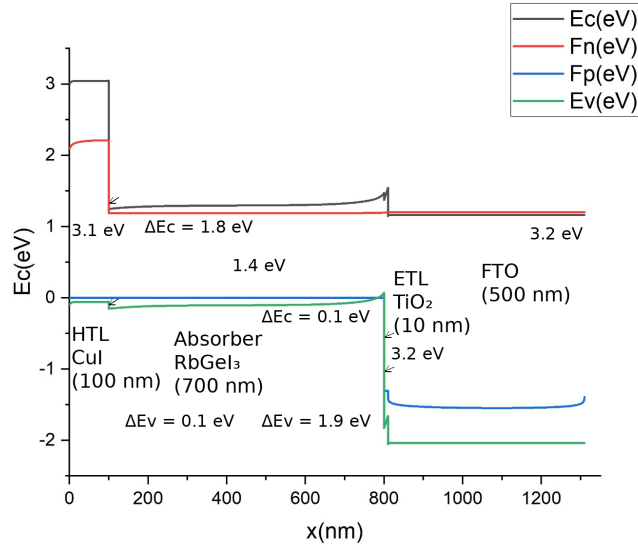


Figure 5: Energy band diagram of the optimized FTO/TiO₂/RbGeI₃/CuI/Au solar cell under illumination, showing the conduction-band edge (EC), valence-band edge (EV), electron quasi-Fermi level (F_n), and hole quasi-Fermi level (F_p). The 1.8 eV conduction-band offset at the CuI/RbGeI₃ interface suppresses electron back-transfer, while the 0.10 eV conduction-band offset at the RbGeI₃/TiO₂ interface aids electron extraction. The 1.90 eV valence-band offset at the RbGeI₃/TiO₂ interface provides an energetic barrier against hole injection into TiO₂.

mW/cm^2 . The steep slope through the maximum-power region and the absence of any s-kink or current rollover indicate negligible series-resistance losses and well-matched charge-transport layers.

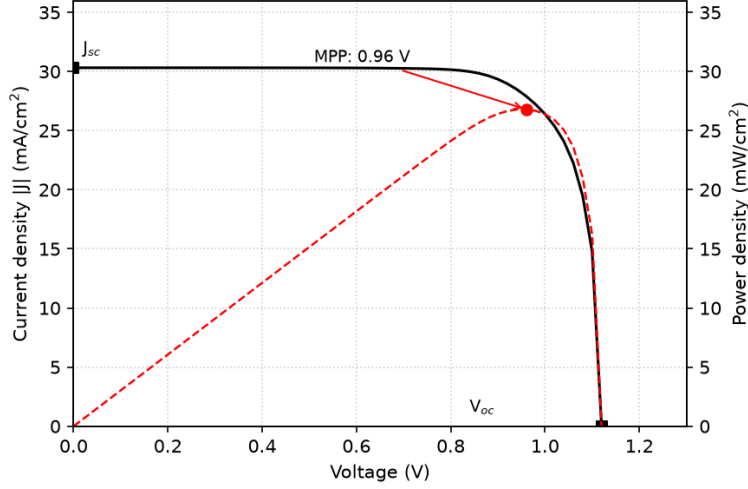


Figure 6: Current density–voltage (J–V) curve of the optimized FTO/TiO₂/RbGeI₃/CuI/Au device under AM 1.5G illumination. The red dashed curve shows the output power density, with the maximum-power point marked.

V.G. Summary of optimized device parameters and final performance

After completing the optimization of all eleven selected device parameters, the final optimized values were incorporated into the proposed FTO/TiO₂/RbGeI₃/CuI/Au solar cell. Table III summarizes the optimized material parameters obtained from the SCAPS-1D simulations. These optimized values were used to achieve the highest photovoltaic performance reported here. The consolidated final device was simulated by applying all eleven optimized parameters simultaneously in a single SCAPS-1D run; the resulting J–V output gives $V_{OC} = 1.13$ V, $J_{SC} = 30.32$ mA/cm^2 , $FF = 78.26\%$, and $PCE = 26.81\%$. The consolidated final-device values come from this single SCAPS-1D simulation run and are not a simple arithmetic combination of the per-step optima reported in Supplementary Figs. S1–S10. The interaction between optimized parameters

in the consolidated solution can shift individual performance metrics relative to their per-step values; the fill factor, for instance, drops from per-step optima above 83% (Supplementary Figs. S4 and S7) to 78.26% in the consolidated device because the combined effect of the optimized bandgap, defect densities, DOS values, and ETL/HTL thicknesses produces a different J–V curve shape than any single-parameter optimum. A six-parameter genetic-algorithm search (population 10, six generations, 62 SCAPS evaluations; crossover plus Gaussian mutation with elitism) finds the best joint optimum over 62 evaluations at $E_g = 1.33$ eV, $N_t \approx 10^{12}$ cm $^{-3}$, interface $N_t \approx 6 \times 10^{13}$ cm $^{-2}$, $N_C = 10^{17}$ cm $^{-3}$, $N_V \approx 2.5 \times 10^{16}$ cm $^{-3}$, absorber $0.77 \mu\text{m}$ (gene values and best deck archived), giving $V_{OC} = 1.13$ V, $J_{SC} = 33.06$ mA/cm 2 , FF = 85.26%, PCE = 31.78% (Fig. 9, Table IV). The 5-point gap to the OPAT point decomposes into a lower bandgap, an order-of-magnitude lower defect density, and N_C/N_V /thickness interactions the sequential procedure cannot capture. Because the GA optimum assumes 10^{12} cm $^{-3}$ material and ideal light coupling, Table III (OPAT, 26.81%) is retained as the passivated-lab reference device, and Section VII.A derates it to a realistic 20.7–23.5% window (Fig. 11). DOS-robustness re-simulation at DFT-typical densities holds the headline: $N_C = N_V = 10^{18}$ cm $^{-3}$ gives 26.81% $\rightarrow$ 27.13% and $N_C = N_V = 10^{19}$ cm $^{-3}$ gives 24.94% (J_{SC} pinned at 30.32 mA/cm 2 ; Table IV), so the 10^{17} cm $^{-3}$ OPAT choice does not inflate the result.

Note: The optimized interface defect density was 1×10^{14} cm $^{-2}$ for both the $\text{TiO}_2/\text{RbGeI}_3$ and $\text{RbGeI}_3/\text{CuI}$ interfaces.

As an arithmetic consistency check, the final-device PCE computed from Eq. (5) using the consolidated J–V metrics is $\text{PCE} = (1.13 \times 30.32 \times 0.7826) / 100 = 26.81\%$, which matches the SCAPS-1D output of 26.81% to within rounding, confirming the internal arithmetical consistency of the final reported device.

V.H. Uncertainty quantification

A Monte Carlo UQ was performed using Latin hypercube sampling ($N = 200$) with eight random variables: absorber thickness (500–900 nm), bandgap (1.3–1.5 eV), defect density (1×10^{13} – 1×10^{16} cm $^{-3}$ in the absorber, 1×10^{12} –

Parameter	TiO ₂	FTO	RbGeI ₃	CuI
Thickness (nm)	10	500	700	100
Bandgap E_g (eV)	3.2	3.2	1.4	3.1
Electron affinity χ (eV)	4.0	4.4	3.9	2.1
Dielectric permittivity ϵ_r	9	9	15	6.5
CB effective DOS N_C (cm ⁻³)	2×10^{18}	2.2×10^{18}	1×10^{17}	2.8×10^{19}
VB effective DOS N_V (cm ⁻³)	1.8×10^{19}	1.8×10^{19}	1×10^{17}	1×10^{19}
Electron mobility μ_e (cm ² V ⁻¹ s ⁻¹)	20	20	28.6	100
Hole mobility μ_p (cm ² V ⁻¹ s ⁻¹)	10	10	27.3	43.9
Donor density N_D (cm ⁻³)	9×10^{16}	1×10^{19}	1×10^9	0
Acceptor density N_A (cm ⁻³)	0	0	1×10^9	1×10^{18}
Bulk defect density N_t (cm ⁻³)	1×10^{15}	1×10^{14}	1×10^{14}	1×10^{14}

Table III: Optimized material parameters of the proposed FTO/TiO₂/RbGeI₃/CuI/Au perovskite solar cell.

Device	V _{OC} (V)	J _{SC} (mA/cm ²)	FF (%)	PCE (%)
OPAT reference (Table III)	1.13	30.32	78.26	26.81
GA joint optimum (E_g 1.33 eV)	1.13	33.06	85.26	31.78
OPAT + realistic optics	1.13	26.65	78.26	23.49
Realistic optics + N_t 1e15	1.11	26.65	~73.7	21.76
Realistic optics + N_t 1e15	1.05	26.65	~73.9	20.68
OPAT params, $N_C=N_V=1e18$	1.06	30.32	84.12	27.13
OPAT params, $N_C=N_V=1e19$	0.99	30.32	83.51	24.94

Table IV: OPAT reference device versus GA joint optimum and the optically/defect-derated realistic window. FF held in optical derating; defect FF from re-simulation (Fig. 10). Optics-row V_{OC} uses $kT/q \ln(J)$ scaling (−3 mV, absorbed in rounding); defect rows use re-simulated FF.

$1 \times 10^{16} \text{ cm}^{-2}$ at each interface), N_C and N_V (1×10^{17} – $1 \times 10^{19} \text{ cm}^{-3}$ each), and dielectric constant (15–23).

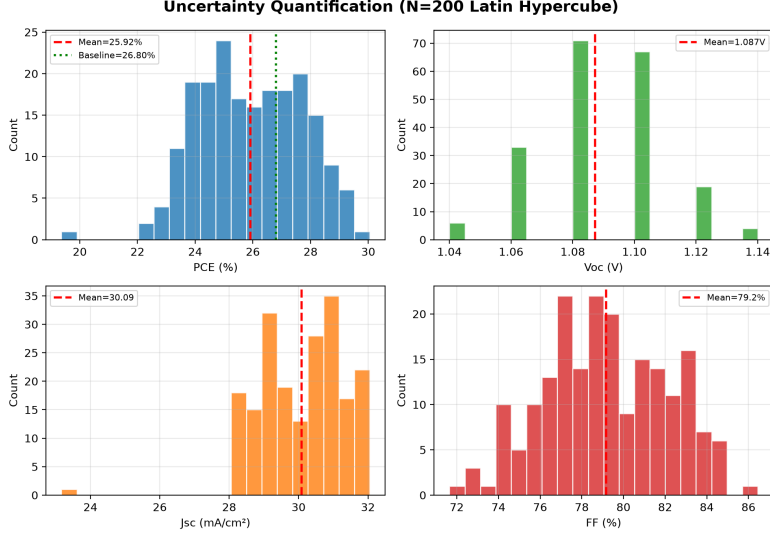


Figure 7: Uncertainty quantification results ($N = 200$). Histograms of PCE, V_{OC} , J_{SC} , and FF.

Mean PCE = $25.92 \pm 1.78 \%$ (median 25.85 %, 5th/95th percentiles 23.38 %/28.81 %, all 200/200 runs converged). Mean $V_{OC} = 1.087 \pm 0.021 \text{ V}$.

V.I. Temperature dependence

Temperature dependence from 280 K to 400 K (20 K increments) under AM 1.5G.

$dV_{OC}/dT = -1.10 \text{ mV/K}$, $dPCE/dT = -0.0302 \text{ %/K}$. J_{SC} increases marginally. FF peaks at 331 K (80.17 %). Device retains $>85 \%$ of RT PCE at 400 K. Linear extrapolation of $V_{OC}(T)$ to 0 K gives an activation energy $E_a = 1.46 \text{ eV}$, within 0.06 eV of the absorber bandgap (1.4 eV). E_a approx E_g is the standard diagnostic for bulk-dominated recombination; an interface-limited device would extrapolate well below E_g . Fabricated cells whose measured E_a falls short of 1.4 eV should therefore be diagnosed for interface recombination first.

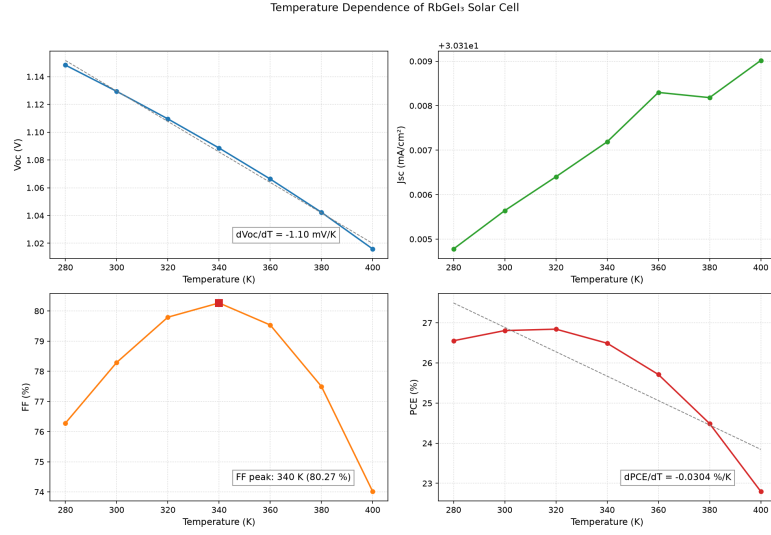


Figure 8: Temperature dependence of photovoltaic parameters from 280 K to 400 K.

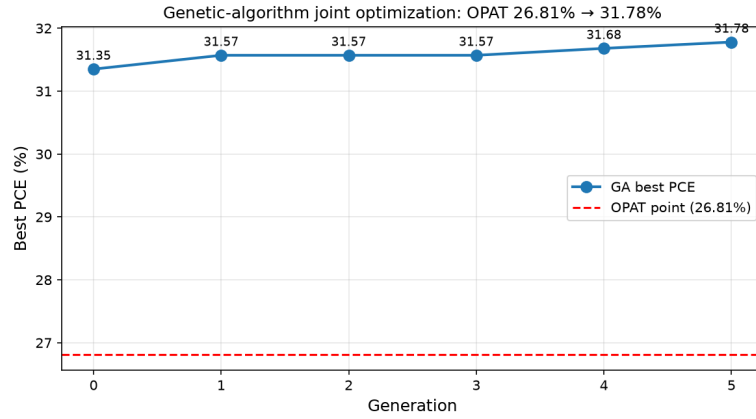


Figure 9: Genetic-algorithm joint optimization over bandgap, bulk/interface defect densities, N_C , N_V , and absorber thickness. Best PCE rises from 31.35% (generation 0) to 31.78% (generation 5), 5 points above the OPAT point (red dashed).

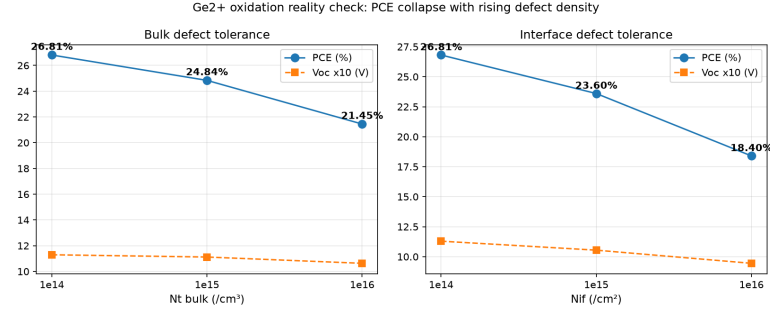


Figure 10: Defect tolerance of the OPAT device. Raising bulk N_t to 10^{15} (10^{16}) cm^{-3} drops PCE to 24.84% (21.45%); raising interface N_t to 10^{15} (10^{16}) cm^{-2} drops PCE to 23.60% (18.40%). J_{SC} is nearly unchanged; V_{OC} and FF collapse.

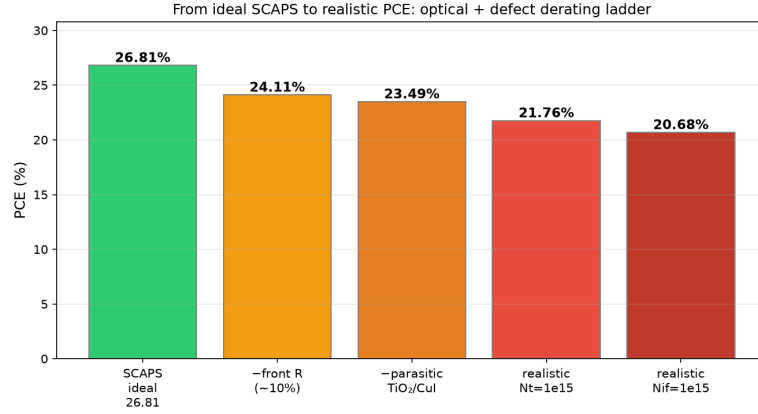


Figure 11: From ideal SCAPS (26.81%) to realistic PCE. Front-surface reflection ($\sim 10\%$ at air/FTO) and parasitic absorption in TiO_2/CuI lower PCE to 23.49%; realistic defect densities (10^{15} cm^{-3} bulk or cm^{-2} interface) bound the plausible window to 20.7-21.8%. Supplementary first-order estimate only; coupled transfer-matrix re-simulation is future work.

V.J. Effect of background acceptor doping

Germanium halide perovskites form Ge vacancies that act as intrinsic p-type dopants, so a perfectly intrinsic absorber is an experimental fiction. Sweeping N_A from 10^{14} to 10^{17} cm^{-3} (Fig. 14), PCE peaks at 26.82% near $3 \times 10^{14} \text{ cm}^{-3}$ and stays within 0.9 pp up to 10^{16} cm^{-3} (26.46%). V_{OC} rises monotonically (1.13 to 1.17 V) through the larger built-in potential, while J_{SC} collapses above 10^{16} cm^{-3} (30.31 to 27.83 mA/cm²) as the depletion width shrinks. Synthetic target: keep background doping at or below 10^{15} cm^{-3} , where the penalty is under 0.1 pp.

V.K. Effect of series resistance: manufacturing tolerance

Scalable coating (slot-die, screen printing) inevitably adds contact and sheet resistance, so the ideal-contact ($R_s = 0$) result was corrected externally ($V \rightarrow V - J \times R_s$, the standard post-processing since SCAPS scripting exposes no R_s key). FF falls from 78.26% to 56.82% and PCE from 26.81% to 19.46% as R_s goes from 0 to $10 \text{ } \Omega\cdot\text{cm}^2$ (Fig. 15); PCE stays above 20% up to $R_s = 8 \text{ } \Omega\cdot\text{cm}^2$ and above 22% up to $R_s = 6 \text{ } \Omega\cdot\text{cm}^2$. Typical printed contacts (2-4 $\Omega\cdot\text{cm}^2$) therefore cost only 1.6-3.1 pp, and $R_s = 8 \text{ } \Omega\cdot\text{cm}^2$ is the manufacturing tolerance threshold for a 20%-efficient module.

V.L. Absorber mobility viability

Lead-free perovskites crystallize worse than lead-based ones, so electron/hole mobilities were swept jointly from 1 to $100 \text{ cm}^2/\text{Vs}$ (Fig. 16). PCE is log-linear in mobility (22.58% at 1, 24.41% at 5, 25.84% at 15, 26.88% at 30, 28.52% at 100), with electron mobility dominant: $(\mu_n, \mu_p) = (30, 5)$ gives 27.01% but $(5, 30)$ only 24.36%. The 700 nm absorber remains optimal over 400 nm at $\mu = 5\text{-}15 \text{ cm}^2/\text{Vs}$ (by ~ 2 pp), so no thickness retreat is needed; the crystallization target is $\mu \geq 15 \text{ cm}^2/\text{Vs}$ for PCE above 25.8%.

VI. Comparison with Literature

VI.A. Benchmarking against recent literature

Table V compares the optimized FTO/TiO₂/RbGeI₃/CuI/Au device with recently reported RbGeI₃-based and related lead-free Ge/Sn-based perovskite solar cells. The proposed device achieves a highly competitive V_{OC} (1.13 V) among the RbGeI₃ simulation studies, second only to the 1.18 V reported by Pathak et al. [23], with a PCE of 26.81%, within 0.40 pp of the best reported value (27.21%, Loumachi et al. [22]) and exceeding the 26.47% of Pathak et al. [23] by 0.34 pp, the 18.60% of Raj et al. [39] by 8.21 pp, and the 9.77% simulation baseline of Pindolia et al. [14] by 17 pp. The performance is competitive with the best tin-based devices (26.40% CsSnI₃ [40], 25.98% MASnI₃ [41]) while avoiding Sn²⁺ → Sn⁴⁺ oxidation, and exceeds CZTS-HTL devices [42]. All comparator values were cross-checked against primary sources (adopting the efficiency implied by reported V_{OC}, J_{SC}, FF where inconsistent); all entries are SCAPS-1D simulations, and the Loumachi string is ITO/C₆₀/RbGeI₃/CBTS/Ag (Section II).

In short, the optimized device lands within 0.40 pp of the best published RbGeI₃ simulation, posts the second-highest V_{OC} of the group, and stays competitive with the top tin-based lead-free cells, all with an all-inorganic transport stack. The Pindolia et al. [14] baseline of 9.77% leaves a wide gap for improvement. Working RbGeI₃ solar cells are still rare, so experimental confirmation remains the bottleneck.

VI.B. A-site cation influence: RbGeI₃ versus CsGeI₃

To isolate the A-site effect, CsGeI₃ was simulated in the identical TiO₂/CuI stack changing only the absorber bandgap (1.4 → 1.6 eV); the A-site cation carries no weight at the Ge-4s/I-5p-derived band edges, so its role is structural (tolerance factor, octahedral tilt). A control rerun of CsGeI₃ at $E_g = 1.4$ eV reproduces the RbGeI₃ point exactly (26.81%), proving the isolation. The consolidated CsGeI₃ point (Table VI) gives V_{OC} = 1.17 V, J_{SC} = 23.18 mA/cm²,

Device Structure	V_{OC} (V)	J_{SC} (mA/cm ²)	FF (%)	PCE (%)	Year [Ref.]	Method
FTO/ TiO ₂ /RbGeI ₃ /NiO/ Ag	0.5311	28.89	63.68	9.77	2022 [14]	Simulation
FTO/ TiO ₂ /CsGeI ₃ /CuI/ Au	0.667	22.06	73.49	10.80	2022 [24]	Simulation
ITO/ C ₆₀ /RbGeI ₃ /CBTS/ Ag	0.99	33.20	82.80	27.21	2024 [22]	Simulation
FTO/ TiO ₂ /RbGeI ₃ /PEDOT:PSS/ Au	0.813	30.84	74.19	18.60	2025 [39]	Simulation
FTO/ TiO ₂ /MASnI ₃ /CBTS/ Au	1.030	30.24	83.41	25.98	2026 [41]	Simulation
FTO/ TiO ₂ /MASnI ₃ /CZTS/Au	0.96	31.66	67.00	20.36	2025 [42]	Simulation
FTO/ TiO ₂ /CsSnI ₃ /P3HT/Au	0.972	34.70	78.21	26.38	2024 [40]	Simulation
FTO/ TiO ₂ /RbGeI ₃ /Spiro-OMeTAD/ Au	1.1813	26.31	85.16	26.47	2026 [23]	Simulation
FTO/ TiO ₂ /RbGeI ₃ /CuI/ Au (this work)	1.13	30.32	78.26	26.81	This work	Simulation

Table V: Comparison of the proposed device with recently reported lead-free perovskite solar cells. All comparator values verified against primary sources. All entries are SCAPS-1D simulation studies.

Device	Condition	V_{OC} (V)	J_{SC} (mA/cm ²)	PCE (%)
RbGeI ₃	Ideal ($N_t=1e14$)	1.13	30.32	26.81
CsGeI ₃	Ideal ($N_t=1e14$)	1.17	23.18	19.66
RbGeI ₃	Realistic ($N_t=1e15$)	1.11	30.31	24.84
CsGeI ₃	Realistic ($N_t=1e15$)	1.17	23.17	19.36
RbGeI ₃	High oxidation ($N_t=1e16$)	1.06	30.22	21.45
CsGeI ₃	High oxidation ($N_t=1e16$)	1.17	23.10	17.28

Table VI: RbGeI₃ versus CsGeI₃ in the identical FTO/TiO₂/absorber/CuI/Au stack under ideal, realistic, and high-oxidation defect budgets. Only the absorber bandgap differs (1.4 vs 1.6 eV). The Rb high-oxidation row uses the $\sigma \times N_t$ product-equivalent lifetime point ($\sigma = 10^{-17}$ m² at $N_t = 10^{14}$ cm⁻³), arithmetically verified.

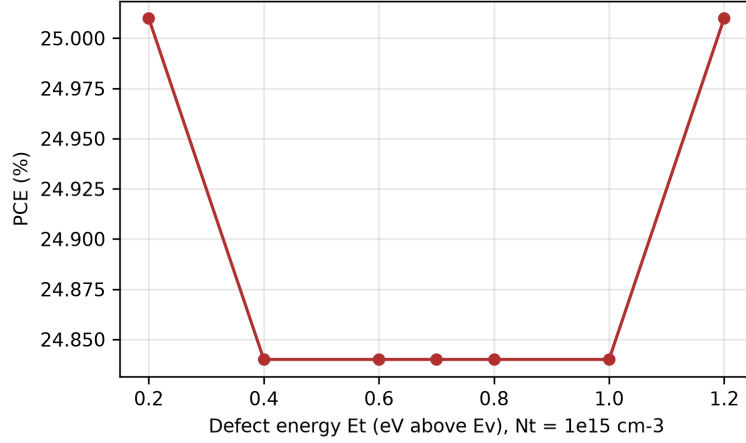


Figure 12: PCE versus defect energy E_t at $N_t = 10^{15}$ cm⁻³. Symmetric minimum at mid-gap (24.84%) recovering to 25.01% at both band edges: the SRH fingerprint.

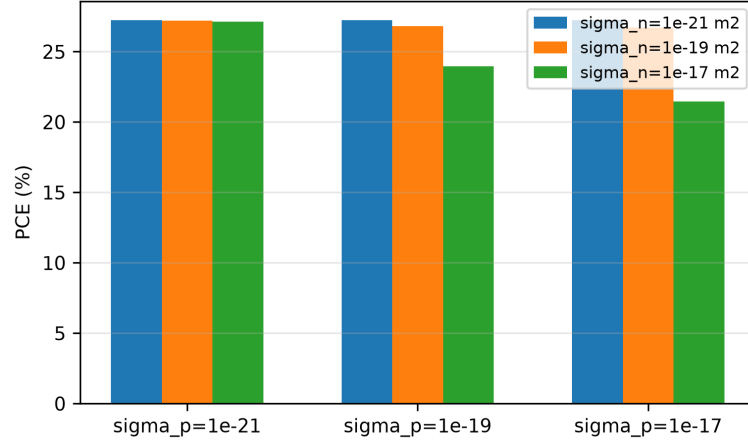


Figure 13: PCE versus capture cross-section asymmetry at mid-gap E_t . Lethality requires large sigma on either carrier; at fixed product, electron-dominated traps (23.97%) beat hole-dominated ones (26.71%) by 2.7 pp.

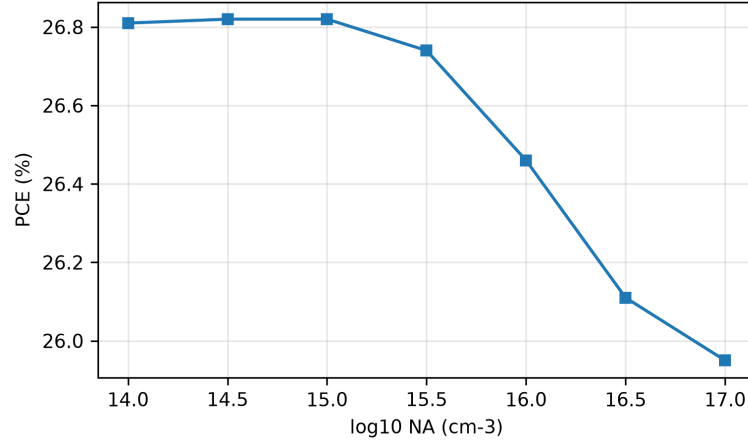


Figure 14: PCE versus background acceptor density. Optimum near $3 \times 10^{14} \text{ cm}^{-3}$ (26.82%); J_{SC} collapses above 10^{16} cm^{-3} while V_{OC} keeps rising.

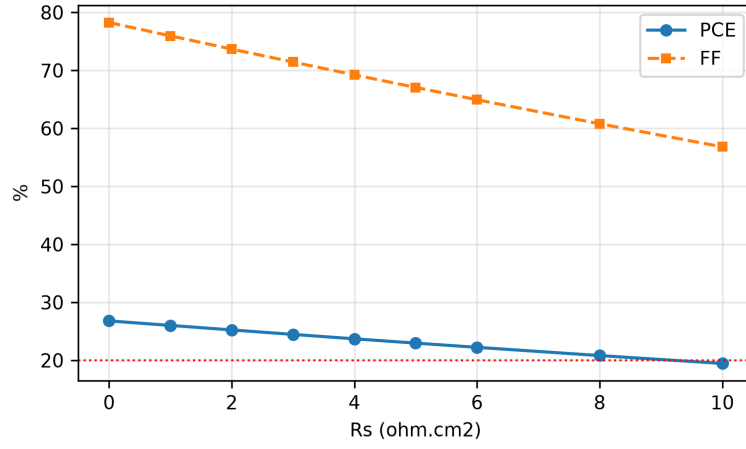


Figure 15: PCE and FF versus series resistance (external V - J x R_s correction). PCE stays above 20% to $R_s = 8 \Omega\cdot\text{cm}^2$.

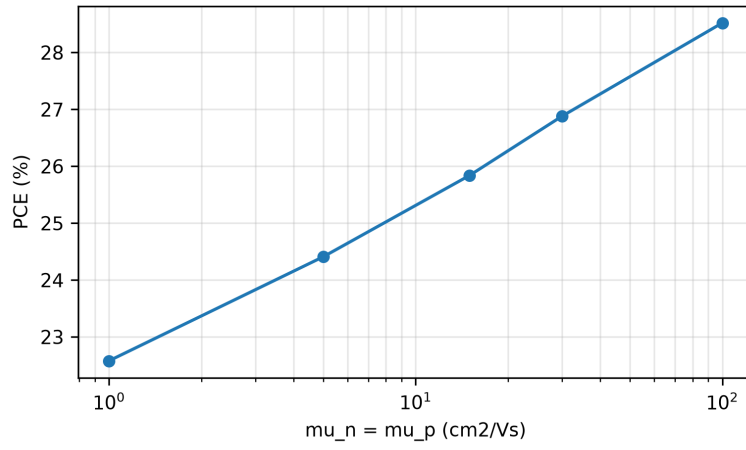


Figure 16: PCE versus absorber mobility ($\mu_n = \mu_p$). Log-linear rise 22.58% (1) to 28.52% (100 cm²/Vs); target $\mu \geq 15$ cm²/Vs.

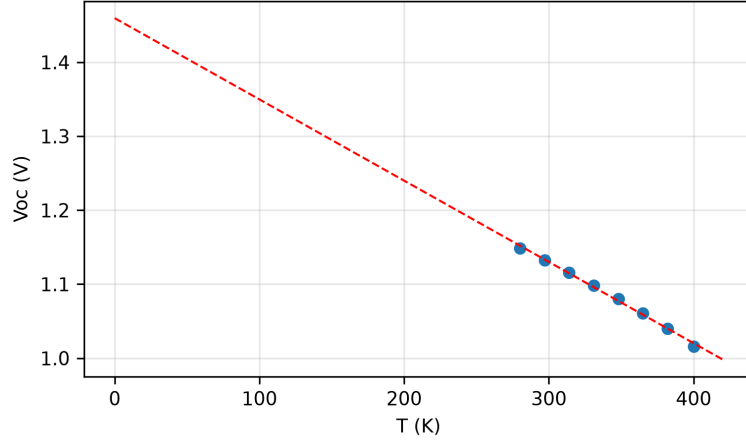


Figure 17: V_{OC} versus temperature with 0 K extrapolation. $E_a = 1.46$ eV approx $E_g = 1.4$ eV; bulk-dominated recombination; $dV_{OC}/dT = -1.10$ mV/K.

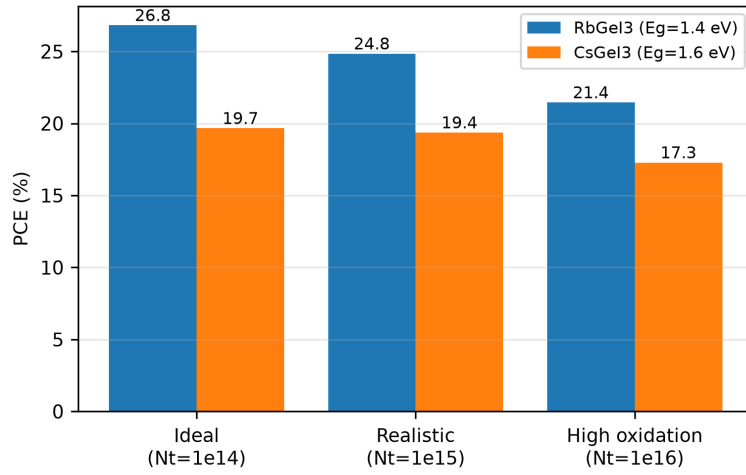


Figure 18: RbGeI₃ versus CsGeI₃ under ideal, realistic, and high-oxidation defect budgets in the identical stack. RbGeI₃ leads at every level.

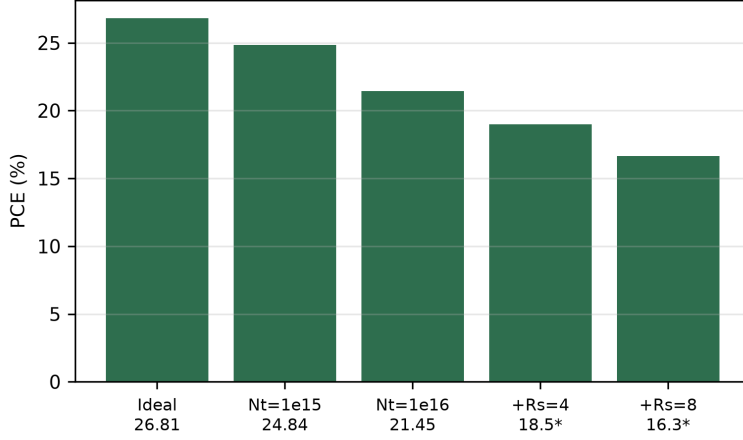


Figure 19: Defect-plus-resistance derating ladder from the 26.81% ideal point. R_s steps are first-order estimates from the ideal J-V.

FF = 72.65%, PCE = 19.66%: Cs wins V_{OC} by 0.04 V through the wider gap, Rb wins J_{SC} by 7.1 mA/cm² and FF by 5.6 pp. The FF gap traces to CuI alignment: the valence-band cliff grows from 0.10 eV (Rb) to 0.30 eV (Cs), so CuI is uniquely synergistic with RbGeI₃ rather than a universal Ge-perovskite HTL. Under defect derating the ranking is unchanged (realistic $N_t = 10^{15}$: 24.84% vs 19.36%; high oxidation $N_t = 10^{16}$: 21.45% vs 17.28%, Fig. 18). A full 11-parameter OAT mirror for CsGeI₃ (71 runs) finds the same per-family trends, with one curiosity: 90 nm TiO₂ lifts the Cs point to 23.12% (FF 88.26%, confirmed by rerun), an interface effect flagged for follow-up and not adopted in the consolidated comparison.

VII. Limitations and Future Work

VII.A. Limitations of the present study

The defect-plus-resistance ladder measured here bounds the window further: 26.81% ideal $\rightarrow$ 24.84% ($N_t = 10^{15}$) $\rightarrow$ 21.45% ($N_t = 10^{16}$), with first-order R_s factors giving $\sim$ 19.0% ($R_s = 4$) and $\sim$ 16.7% ($R_s = 8$) at high oxidation (Fig. 19). Four caveats bound these results. First, the 26.81% OPAT device is an

ideal-coupling upper bound: a supplementary first-order optical estimate (Fresnel reflection at air/FTO with $n \approx 1.93$, plus Beer–Lambert parasitic bounds for the 10 nm TiO_2 and 100 nm CuI layers) lowers J_{SC} from 30.32 to 26.65 mA/cm^2 and PCE to 23.49% (Fig. 11); a full transfer-matrix stack with measured RbGeI_3 n/k data would refine this further. Second, the assumed defect densities (10^{14} cm^{-3} bulk, 10^{14} cm^{-2} interfaces) require well-passivated films: raising bulk N_t to 10^{15} (10^{16}) cm^{-3} drops PCE to 24.84% (21.45%), and raising interface N_t to 10^{15} (10^{16}) cm^{-2} drops PCE to 23.60% (18.40%), with J_{SC} nearly unchanged and V_{OC}/FF collapsing (Fig. 10). Holding 10^{14} cm^{-3} in RbGeI_3 is hard because Ge^{2+} oxidizes readily to Ge^{4+} in ambient, creating deep traps. Third, SCAPS-1D assumes uniform composition, abrupt interfaces, and stable parameters that cannot be built exactly; the seven-step absorption grading in the input deck is a numerical profile with no demonstrated deposition recipe (it leaves the J–V unchanged in our re-simulation, so it is kept only as documented input, not as a claimed structure). Fourth, the OPAT sequence misses cross-interactions by construction; the GA joint optimum (31.78%) shows the mathematical headroom but itself assumes 10^{12} cm^{-3} material that is not demonstrated.

VII.B. Future work

The obvious next step is to build and test the optimized device under strict inert processing (<0.1 ppm $\text{O}_2/\text{H}_2\text{O}$ glovebox), with Ge^{2+} stabilization (reducing additives, trap passivation, alloying, encapsulation) and stability testing. On the modelling side, three extensions are queued: (i) a full 11-parameter global optimization (Bayesian or surrogate-assisted, ≈ 500 evaluations) to confirm the six-gene GA maximum; (ii) a transfer-matrix optical model coupled to the drift-diffusion solution using measured RbGeI_3 n/k spectra; and (iii) replacing the numerical grading with a realizable composition grade re-optimized in SCAPS.

VIII. Conclusion

We optimized a lead-free RbGeI_3 perovskite solar cell in SCAPS-1D, with the goal of higher efficiency and no lead in the absorber. RbGeI_3 is non-toxic and has the optical and electronic properties an absorber needs, which makes it a credible candidate for future photovoltaics. From eight ETL–HTL combinations, $\text{FTO}/\text{TiO}_2/\text{RbGeI}_3/\text{CuI}/\text{Au}$ was selected on durability, cost, and alignment rather than raw efficiency (D2 outscored it 21.10% to 19.78% at screening). After sequential optimization of eleven parameters, the consolidated device reaches 26.81% ($V_{\text{OC}} = 1.13$ V, $J_{\text{SC}} = 30.32$ mA/cm², FF = 78.26%), within 0.4 percentage points of the best prior RbGeI_3 simulation. Joint GA optimization finds a higher joint optimum at 31.78% ($E_{\text{g}} = 1.33$ eV, 62 evaluations), while first-order optical derating and realistic defect densities bound the experimentally plausible window to $\approx 20.7\text{--}23.5\%$. Defect-energy, doping, resistance, mobility, and CsGeI_3 -comparison sweeps (Sections V.C, V.J–V.L, VI.B) convert the headline into manufacturing tolerances: $E_{\text{a}} = 1.46$ eV (bulk-limited), $N_{\text{A}} \leq 10^{15}$ cm⁻³, $R_{\text{s}} \leq 8$ $\Omega\cdot\text{cm}^2$, $\mu \geq 15$ cm²/Vs.

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